TEXORA · LECTURE SHEET

Carding

The heart of the spinning mill: between wire-covered surfaces moving at different speeds, the card separates a matted lap fibre from fibre, removes trash, neps and short fibre, and delivers the first true sliver.

TE-2101 · SPINNING TECHNOLOGY-I · TOPIC 3 · LEVEL 2 TERM 1

1. Objects of carding

Blow-room opening leaves the cotton as small tufts still holding trash, seed fragments, neps (tiny entangled fibre knots) and short fibre mixed with the spinnable long fibre. No drafting system downstream can separate fibres still locked in tufts, so carding is the last and most thorough cleaning and separation stage before sliver formation. Its objects are:

- **Individualisation of fibres** — to open tufts progressively until, ideally, every fibre is separate, so later drafting can move fibres individually and evenly.
- **Removal of trash and dust** — remaining leaf, seed coat fragments and dust are thrown out by centrifugal action and air currents, mainly under the licker-in and cylinder.
- **Removal of neps** — neps are either opened out by the intense cylinder–flat working, or carried away in the flat strips; a well-set card is the mill's principal nep-removal machine.
- **Removal of short fibre** — fibres too short to control are taken away in the flat strips and droppings, slightly raising the mean length of what goes forward.
- **Partial straightening and parallelisation** — fibres leave straighter and more parallel than they entered, though with hooks (mainly trailing hooks) for the draw frame to remove later.
- **Sliver formation** — to condense the web stripped from the doffer into a continuous, coiled **sliver** of defined weight per unit length, the standard form for all later spinning preparation.

Because carding does so much in one machine, its condition dominates yarn quality: a fault made at the card — nep generation, fibre breakage, uneven feeding — cannot be fully repaired downstream. Hence the old mill saying: the card is *the heart of the spinning mill*.

2. Passage of material through the card

Follow one element of cotton from the back of the machine to the can (Figure 1):

1. **Lap.** The scutcher lap (about 350–450 g/m), wound on a lap rod, unrolls slowly and evenly from the lap stand.
2. **Feed roller and feed plate.** The fluted, spring-loaded **feed roller** grips the lap against the curved nose of the **feed plate** and advances it slowly. Fibres are held at one end while the licker-in strikes the other — the first working zone.
3. **Licker-in (taker-in).** A small roller (about 230–250 mm diameter) clothed with saw-tooth wire, rotating fast (about 800–1,200 rpm). Its teeth tear the held fringe into tiny tufts and hurl trash and heavy impurities through the grid bars beneath it by centrifugal force; mote knives set close to its surface help strip out seed fragments and leaf.
4. **Cylinder (swift).** The large main roller (about 1,250–1,290 mm diameter, surface speed far higher than the licker-in) strips the tufts off the licker-in at the transfer point (Section 4) and carries them into the main carding zone under the flats.
5. **Revolving flats.** An endless chain of about 100–116 narrow bars, each clothed with flexible wire fillet, travels slowly over the cylinder; about 40–46 flats are in working contact at any moment. Between flats and cylinder the decisive fibre-by-fibre separation — the **carding action** — takes place.
6. **Doffer.** A smaller wire-clothed roller (about 600–700 mm diameter) set very close to the cylinder. The cylinder *condenses* its fibre onto the slower doffer surface, which carries it round as an almost transparent film — the **web**.
7. **Web and stripping.** A fast oscillating or rotary doffer comb (or stripping/crush rollers on modern cards) strips the web cleanly from the doffer clothing.
8. **Trumpet and calendar rollers.** The web is gathered through a **trumpet**, which condenses it into a round sliver, and a pair of **calendar (calender) rollers** compresses the sliver, squeezes out air and gives it cohesion.
9. **Coiler.** A rotating coiler head lays the sliver in overlapping coils inside a revolving can, so a large length is stored without tangling and can be withdrawn from the top without stretching.

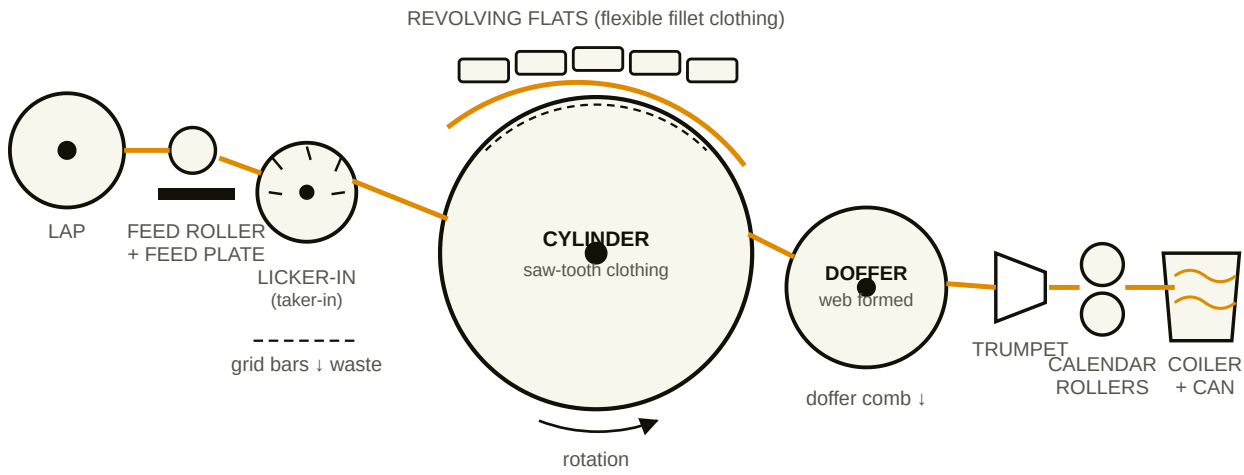


Figure 1 — Passage of material through a revolving-flat card, side view (not to scale). Waste falls out below the licker-in and cylinder; fibre goes forward on the cylinder surface.

3. Card clothing — the working skin of the card

Every working surface is covered with **card clothing**: a dense field of short wire points that grip, tease and transfer the fibres. Two families are used:

3.1 Saw-tooth (metallic) wire

A continuous wire of saw-tooth cross-section, wound in a close spiral into grooves on the roller surface and ground to sharp points, used on the **licker-in, cylinder and doffer**. It is rigid, stays sharp, resists loading with fibre, allows high speeds and production, and needs grinding only at long intervals. Tooth geometry — height, pitch (points per unit area) and front angle — is chosen per organ: fine and dense on the cylinder, to hold a very thin fibre film against the flats; shaped on the doffer to *receive and hold* condensed fibre and then release it cleanly; coarse and strongly inclined on the licker-in, to strike, tear tufts and throw off trash.

3.2 Flexible fillet clothing

Short U-shaped wire hooks set through a foundation of woven fabric and leather/rubber layers, wound as a **fillet** onto the flat bars and used on the **revolving flats**. The points are resilient: they yield slightly on contact, giving the gentle, progressive teasing wanted opposite the rigid cylinder — the cylinder–flat pair is deliberately one rigid surface working against one flexible surface. Flexible points wear dull in service, so flats are ground and periodically re-clothed; worn flat clothing is a direct cause of rising nep count.

FEATURE	SAW-TOOTH METALLIC CLOTHING	FLEXIBLE FILLET CLOTHING

Used on	Licker-in, cylinder, doffer	Revolving flats
Point form	Teeth in a continuous wire, wound on the roller	Hooked staples set in a fabric foundation
Behaviour	Rigid, aggressive	Resilient, yields on contact, gentler
Maintenance	Light grinding at long intervals	Regular grinding; re-clothing when hooks wear
Failure if neglected	Loaded clothing, poor transfer, rising waste	Dull points roll fibres into neps

4. Carding action and stripping action

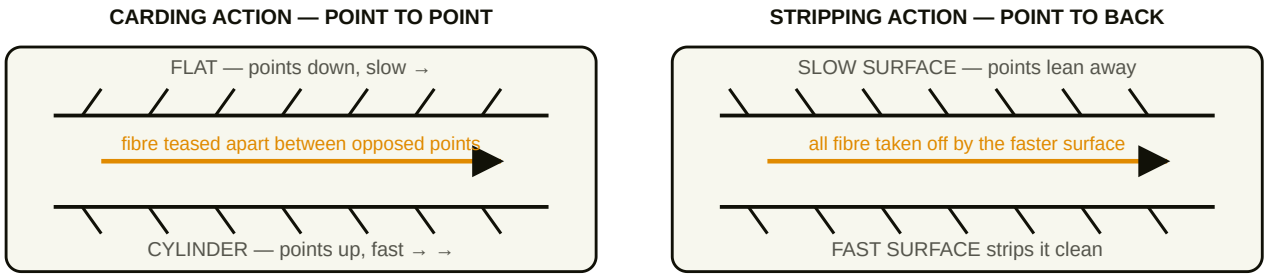
Wherever two clothed surfaces pass close together, one of two fundamental actions occurs — decided by the **direction of the wire points relative to each other** and by the **relative surface speeds**.

4.1 Carding action — point to point

When the points of the two surfaces face *toward each other* (point to point) and one surface moves faster, neither can simply take all the fibre: each point field holds back the fibre it has caught while the opposing points pull at it. Tufts are therefore **teased apart progressively**, fibre by fibre — this is the **carding action**. The classic example is **cylinder–flat**: the fast cylinder carries fibre under the slow flats, points opposed, and in that gap tufts are opened to single fibres, neps are worked open or extracted, and short fibre and dirt pass into the flat clothing, to be stripped away later as flat strips. A milder carding action occurs between feed roller/plate and licker-in, where held fibre is struck by the licker-in teeth.

4.2 Stripping action — point to back

When the points of one surface face the *backs* of the other's points (point to back) and the receiving surface moves faster, the faster surface **strips** — takes all the fibre off — the slower one: a clean transfer, with no teasing. Examples: **licker-in → cylinder** (the much faster cylinder strips the licker-in completely, so no fibre returns under the machine), and the stripping brush that clears waste from the returning flats. The cylinder–doffer transfer is distinct: the doffer is *slower*, points are opposed, and fibre **condenses** from the fast, thinly covered cylinder onto the doffer — which is why the doffer web is denser than the cylinder film.



Left: cylinder–flat (and feed–licker-in) — fibre is worked and opened. Right: licker-in → cylinder — fibre is transferred completely, with no working.

Figure 2 — The two fundamental actions between clothed surfaces. Point direction plus relative speed decides whether fibre is carded (opened) or stripped (transferred).

5. Card settings and their effects

A **setting** is the measured gap between two working surfaces, checked with feeler gauges with the machine stopped. Card settings are extraordinarily close — tenths of a millimetre — because the fibre layer is only a few fibres thick. Closing a setting intensifies the action (more opening, cleaning, transfer) but risks fibre breakage, clothing damage and nep generation if overdone; opening it does the reverse.

SETTING	TYPICAL ORDER	IF SET CLOSER	IF SET WIDER
Feed plate nose – licker-in	About 0.12–0.25 mm	Better opening, more trash out — but fibre rupture and good fibre in the droppings if excessive	Tufts torn off in larger, poorly opened pieces; trash carried forward
Licker-in – cylinder	About 0.12–0.18 mm	Cleaner, more complete stripping transfer	Untransferred fibre is carried round and thrown into the droppings — waste rises, transfer patchy
Cylinder – revolving flats	About 0.20–0.30 mm (the 0.25 mm class)	Most intense carding action: best individualisation and nep removal — the key quality setting; too close risks fibre breakage and clothing contact	Weak carding: tufts survive, neps pass forward, quality falls
Cylinder – doffer	About 0.10–0.13 mm — the closest setting on the card	More complete condensation to the doffer; clearer web	Untransferred fibre recirculates, loads the clothing, web becomes patchy
Mote knives / cylinder	Fraction of a millimetre	Knives strip seed and leaf from the licker-in; plates control air flow over	Wrong settings raise fly, waste and web breaks

plates	the cylinder

Two practical rules follow. First, check settings across the *whole width*: a cylinder 0.25 mm from the flats at the centre and 0.40 mm at one side cards unevenly across its width, and later doubling cannot fully hide it. Second, settings interact with **speed and production rate**: a higher feed thickens the fibre film, so a setting correct at lower production may need review — quality is a balance of setting, speed and load.

6. Nep removal and card waste

6.1 How neps are removed

A nep reaches the card as a pre-formed knot from ginning and the blow room — or is *created* at the card if dull clothing rolls fibres instead of opening them. Between cylinder and flats it receives repeated point-to-point working: many neps are **opened out** into separate fibres, which is genuine removal; the rest are caught by the flat clothing (the slow surface collects what the cylinder cannot open) and leave in the **flat strips**. Nep removal thus depends on sharp clothing, a correct cylinder–flat setting, a moderate fibre load, and flats stripped clean on every return — loaded flats re-deposit their neps. Immature fibres nep most readily, so raw material sets the limit.

6.2 Types of card waste

WASTE	WHERE IT LEAVES	WHAT IT CONTAINS	CONTROL
Licker-in droppings	Below the licker-in, through the grid bars	Heaviest trash: seed fragments, leaf, dust — plus some spinnable fibre torn out with it	Feed plate setting, mote knives, grid bars; excessive droppings = good fibre lost
Flat strips	Stripped from the flats on their return	Short fibre, neps, fine trash and dust from the carding zone — the card's quality waste	Cylinder–flat setting and clothing condition; strips thin and even across the width
Stripping waste, fly, sweepings	Periodic clothing cleaning; fly around the machine	Fibre fly, dust, short fibre	Clothing condition, plate settings, cleaning routine

Total card waste for cotton is commonly about 3–7 % of the feed. Waste is not judged by percentage alone: low waste achieved by passing trash and neps forward is false economy, and high waste achieved by throwing out spinnable fibre is direct money loss — so the mill examines waste *composition* (how much good fibre the droppings and strips contain) as carefully as its weight.

WORKED EXAMPLE — CARD PRODUCTION

A card delivers sliver from the doffer side at a delivery (calendar roller) speed of 150 m/min, and the sliver weight is 5 g/m. Find the production per hour.

Length delivered in one hour = 150 m/min × 60 min = 9,000 m.

Mass delivered = 9,000 m × 5 g/m = 45,000 g = 45 kg.

Production = 150 × 60 × 5 ÷ 1000 = **45 kg/h** (theoretical; at 95 % efficiency, actual output = 45 × 0.95 ≈ 42.8 kg/h).

WORKED EXAMPLE — CARD DRAFT

The card is fed with lap of 400 g/m and delivers sliver of 5 g/m. Find the (mechanical) draft of the card.

Draft = feed weight per unit length ÷ delivery weight per unit length = 400 ÷ 5 = **80**.

Check by lengths: 1 m of lap (400 g) becomes 400 ÷ 5 = 80 m of sliver (80 × 5 g = 400 g, ignoring the small part lost as waste, so this weight-based draft is the nominal draft).

$$\text{Draft} = (\text{feed weight per unit length}) \div (\text{delivery weight per unit length})$$

$$\text{Production (kg/h)} = \text{delivery speed (m/min)} \times 60 \times \text{sliver weight (g/m)} \div 1000$$

The card's draft of roughly 80–110 is the largest single attenuation before the draw frame — achieved not by roller drafting but by spreading the lap on the cylinder into a thin film and re-condensing it on the doffer, which is why it does not damage evenness as an equivalent roller draft would.

Glossary

Carding action — teasing apart of tufts between point-to-point clothed surfaces at different speeds. **Stripping action** — complete transfer of fibre from a slower to a faster surface whose points face the backs of the first surface's points. **Licker-in (taker-in)** — the fast saw-tooth roller tearing the lap fringe into small tufts and ejecting heavy trash. **Cylinder (swift)** — the large main roller carrying fibre through the carding zone. **Revolving flats** — the endless chain of fillet-clothed bars working point-to-point against the cylinder. **Doffer** — the roller onto which the cylinder condenses its fibre film as a web. **Clothing** — the wire point covering (saw-tooth or flexible fillet) of a carding surface. **Nep** — a tiny entangled knot of fibres. **Flat strips** — waste stripped from the flats: short fibre, neps, fine trash. **Licker-in droppings** — heavy waste ejected below the licker-in. **Setting** — the measured gap between two working surfaces. **Sliver** — the continuous untwisted rope of fibres the card delivers.

Key points

• The card individualises fibres, removes trash, neps and short fibre, and delivers the first sliver — the heart of the spinning mill.
• Passage: lap → feed roller and feed plate → licker-in → cylinder → revolving flats → doffer → web → trumpet → calendar rollers → coiler and can.
• Saw-tooth metallic clothing covers licker-in, cylinder and doffer; flexible fillet clothing covers the flats.
• Point-to-point at different speeds gives carding action (cylinder–flat); point-to-back with a faster receiver gives stripping action (licker-in → cylinder).
• The cylinder–flat setting (0.25 mm class) is the key quality setting; cylinder–doffer is the closest setting on the machine.
• Card waste is mainly licker-in droppings (heavy trash) and flat strips (short fibre, neps) — judge it by good-fibre content, not weight alone.
• Card draft is about 80–110 by weight (400 g/m lap → 5 g/m sliver gives 80).

Practice

1. A card delivers sliver at 120 m/min with a sliver weight of 4.5 g/m and runs at 92 % efficiency. Calculate the actual production in kg/h, showing each step.
2. A card is fed with lap of 380 g/m and delivers sliver of 4.75 g/m. Calculate the card draft and check your answer by a length argument.
3. At the cylinder–flat setting, a technician opens the gap well beyond the 0.25 mm class to "be safe" against clothing contact. State three consequences for sliver and yarn, explaining each.
4. Distinguish carding action from stripping action by point direction and relative speed, naming one machine pair showing each.
5. A card's licker-in droppings suddenly contain much more white, spinnable fibre, and total waste rises. Give three setting or condition faults that could cause this, with the check for each.
6. Why is the card, not the blow room, the principal nep-removal machine? Answer in terms of what happens to a nep between cylinder and flats, and where unopened neps leave.

Answer key

1. Length per hour = 120 × 60 = 7,200 m. Mass = 7,200 × 4.5 = 32,400 g = 32.4 kg/h theoretical. Actual = 32.4 × 0.92 = 29.808 ≈ **29.8 kg/h**.
2. Draft = 380 ÷ 4.75 = **80**. Check: 380 g of lap at 4.75 g/m makes 380 ÷ 4.75 = 80 m of sliver (80 × 4.75 = 380 g, ignoring waste) — consistent.
3. (i) Tufts are not fully opened — they pass into the sliver as thick places and cause uneven drafting later. (ii) Nep removal falls — neps are neither opened nor taken into the flats, so sliver and yarn nep

count rises. (iii) Short fibre and fine trash are less effectively extracted to the flat strips, lowering cleanliness and yarn appearance.

4. Carding action: points face each other (point to point), surfaces at different speeds; each holds fibre while the other pulls, so tufts are teased apart — e.g. cylinder–flats. Stripping action: the receiver's points face the backs of the donor's points (point to back) and the receiver is faster, taking all the fibre in a clean transfer — e.g. licker-in → cylinder.
5. Any three: feed plate nose too close to the licker-in, so the fringe is struck too violently (check with feeler gauges); licker-in–cylinder setting too wide, so untransferred fibre returns and is thrown out (check the setting); mote knives too close or grid bars too open, letting fibre fall with the trash (check knife/bar settings); licker-in clothing damaged, or feed rate raised beyond its capacity (inspect clothing and lap weight).
6. Between cylinder and flats each nep receives repeated point-to-point working: most are teased open into separate fibres — true removal no blow-room beater can perform on such small knots — and the rest are held by the slow flat clothing and leave in the flat strips. The blow room removes only loose trash around tufts.
